

Seminar in Hodge Theory 7/5 中橋 一朗 Hajime Nakahashi

§4 Hodge structure I : Hodge structures and polarisations.

Ref: - Voisin "Hodge theory and complex Algebraic geometry I" Ch 7
 - Barth, Hulek, et al "Compact complex surfaces" § 13, 14
 - Huybrechts "Lectures on K3 surfaces" ch 3
 - Peters, Steenbrink "Mixed Hodge structures" ch 2

Contents

- §1 Definitions and Basic properties.
- §2 Compact Kähler mfd case.
- §3. Polarisation.
- §4. Geometry of Hodge str of weight 1 and 2.
- §5 Functoriality.

§ Definitions and Basic properties.

Let $V_{\mathbb{Z}}$: free $\mathbb{Z}$ -module of finite type.

$V_{\mathbb{Q}}$: real vect. sp. obtained by scalar extension.

$V_{\mathbb{C}}$: complex

Here, note that complex conjugation $z \mapsto \bar{z}$ is well-defined on $V_{\mathbb{C}}$.

Def An integral Hodge str of weight k on $V_{\mathbb{Z}}$ is given by a direct sum decomp. of the complex vect. sp $V_{\mathbb{C}}$.

$$V_{\mathbb{C}} := V_{\mathbb{Z}} \otimes \mathbb{C} = \bigoplus_{p+q=k} V^{p,q}$$

satisfying $V^{p,q} = \overline{V^{q,p}}$

• Rational HS on $V_{\mathbb{Q}}$ is defined in same way.

• $h^{p,q}(V) := \dim V^{p,q}$: Hodge numbers.

• For even weight ($n=2k$), $V \cap V^{k,k}$: Hodge class in V ($=V_{\mathbb{Z}} \text{ or } V_{\mathbb{Q}}$)

$(\mathbb{Z}-, \mathbb{Q}-)$

- Sub-Hodge str. of H.S. V of weight n is give by a $\mathbb{Z}$ - $(\mathbb{Q}-)$ linear subspace $V' \subset V$ such that, H.S. on V induces a H.S. on V' i.e $V_{\mathbb{C}}' = \bigoplus (V_{\mathbb{C}}' \cap V^{p,q})$

• $(\mathbb{Z}-)$ primitive A sub-H.S. $V' \subset V_{\mathbb{Z}}$ is called primitive if V/V' is torsion free.

• $(\mathbb{Z}-)$ irreducible Any Hodge str. that does not contain any non-triv. proper and primitive H.S. $V' \subset V$ is called irreducible.

① Most of the standard linear algebra constructions have analogues in Hodge theory. (c.f. [Huy] p.36-)

i) Direct sum (V, W : same weight)

$$(V+W)^{p,q} = V^{p,q} \oplus W^{p,q}$$

ii) tensor product (V, W : weight n & m resp.)

$V \otimes W$: weight $n+m$

$$(V \otimes W)^{p,q} = \bigoplus V^{p_1, q_1} \otimes W^{p_2, q_2}$$

all pairs of tuples $(p_1, q_1), (p_2, q_2)$ with $p_1 + p_2 = p, q_1 + q_2 = q$.

iii) dual

$V_{\mathbb{Z}}$: integral weight n .

$V^* := \text{Hom}_{\mathbb{Z}}(V, \mathbb{Z})$ by weight $-n$.

$$(V^*)^{-p, -q} := \text{Hom}_{\mathbb{C}}(V^{p, q}, \mathbb{C}) \subset \text{Hom}_{\mathbb{C}}(V_{\mathbb{C}}, \mathbb{C}) = V_{\mathbb{C}}^*$$

Example • Hodge structure of Tate $\mathbb{Z}(1)$

$2\pi i \mathbb{Z} \subset \mathbb{C}$ rank 1 where $\mathbb{Z}(1)^{-1, -1}$ is one-dim.

⚠ why we need this is not apparent here.

See Alessio Corti "Intro to mix Hodge structure".

(In fact, difference between just considering the free module $\mathbb{Z}$ is not important.)

$$\mathbb{Z}(-1) := \mathbb{Z}(1)^* \text{ weight } 2.$$

$$\mathbb{Z}(k) := \mathbb{Z}(1)^{\otimes k} \text{ weight } -2k \text{ for } k > 0. \text{ (underlying } \mathbb{Z}\text{-module } (2\pi i)^k \mathbb{Z})$$

iv) twist $V(k) := V \otimes \mathbb{Z}(k)$ weight $n-2k$.
 where $V(k)^{p,q} = V^{p+k, q+k}$.

v) exterior product

skip see [Huy]. P.37.

Filtration

Given H.S. of weight n , we define the associated Hodge Filter $F^\bullet V$ by

$$F^p V_{\mathbb{C}} = \bigoplus_{r \geq p} V^{r, n-r}.$$

This is a decreasing filtration on $V_{\mathbb{C}}$.
 $0 \subset F^n V_{\mathbb{C}} \subset F^{n-1} V_{\mathbb{C}} \subset \dots \subset F^0 V_{\mathbb{C}} \subset V_{\mathbb{C}}.$

$$\text{Then } F^p V_{\mathbb{C}} \oplus \overline{F^{n-p+1} V_{\mathbb{C}}} = V_{\mathbb{C}} \quad (V^p) \quad (*)$$

Note that the Hodge filtration w/ (*) determines the Hodge decomposition by $V^{p,q} = F^p V_{\mathbb{C}} \cap \overline{F^q V_{\mathbb{C}}}$

- The two notions describe the same structure.
- However, Hodge filtration is more natural when it comes to deforming Hodge structure.

§2 Case of cpt Kähler mfd.

X : cpt Kähler mfd w : Kähler class

$H^k(X, \mathbb{Z})/\text{torsion}$: torsion free part of the singular cohomology
(In what follows, we will often use the same notation modulo torsion & integral cohomology)

A natural H.S. of weight n given by standard Hodge decomposition.

$$H^n(X, \mathbb{Z}) \otimes \mathbb{C} = H^n(X, \mathbb{C}) = \bigoplus_{p+q=n} H^{p,q}(X).$$

Here $H^{p,q}(X)$ = Doubeault cohomology $H^q(X, \Omega^p)$.

This looks like it depends on the choice of w .

Prop 7.5

$FPA^k(X)$: the set of complex diff. forms which are sums of type $(r, k-r)$ with $r \geq p$ at every point.

Then

$$FPH^k(X, \mathbb{C}) = \frac{\ker(d: FPA^k \rightarrow FPA^{k+1})}{\text{Im}(d: FPA^{k-1} \rightarrow FPA^k)}$$

pf see text. ✱

This then tells us the uniqueness of Hodge decomposition.

Remark $\bigoplus H^{2k}(X, \mathbb{Q})$: even part contains all alg. classes i.e. fundamental class $[Z]$ of sub variety $Z \subset X$.

Moreover we can see that $\begin{cases} [Z] \text{ is integral class} \\ [Z] \text{ is in Hodge class } H^{k,k}(X) \end{cases}$

Hodge conjecture asserts that the space spanned by those is determined entirely by the H.S. itself.

Conj X : sm proj var / $\mathbb{C}$.

$$H^{2k}(X, \mathbb{Q}) \cap H^{k,k}(X) = \langle [Z] \mid Z \subset X \rangle_{\mathbb{Q}}$$

§3 Polarisation.

Motivation

X : n -dim cpt Kähler mfd. w/ Kähler form $\omega \in H^2(X, \mathbb{R})$

The cup product $L: H^k(X, \mathbb{R}) \rightarrow H^{k+2}(X, \mathbb{R})$ with $[\omega] \in H^2(X, \mathbb{R})$ gives Lefschetz decomp.

$$H^k(X, \mathbb{R}) = \bigoplus_r L^r H_{\text{prim}}^{k-2r}$$

where $H^k(X, \mathbb{R})_{\text{prim}} = \ker(L^{n-k+1})$ for $k \leq n$

Here each component admits an induced Hodge decomp. since L is of bidegree $(1,1)$

Moreover L gives an intersection form on $H^k(X, \mathbb{R})$ for $k \leq n$.
 $Q(\alpha, \beta) = \int_X \omega^{n-k} \wedge \alpha \wedge \beta = \langle L^{n-k} \alpha, \beta \rangle$

Note that Q is $\begin{cases} \text{alternating} & \text{if } k \text{ is odd} \\ \text{symmetric} & \text{if } k \text{ is even} \end{cases}$

The induced Hermitian form

$H(\alpha, \beta) = i^k Q(\alpha, \bar{\beta})$ satisfying following properties

- $$\left\{ \begin{array}{l} \text{i) The Hodge decomp is orthogonal for } H \\ \text{ i.e. } H(H^{p,q}, H^{s,r}) = 0 \text{ for } (p,q) \neq (s,r) \\ (\Leftrightarrow \text{i}') H(F^m, F^{k-m+1}) = 0 \\ \text{ii) } i^{p-q-k} (-1)^{\frac{k(k-1)}{2}} H(\alpha, \alpha) > 0 \text{ for } \alpha \neq 0 \in H^{p,q} \end{array} \right.$$

Note that Lefschetz decomp. is NOT decomp as a direct sum of rational sub-Hodge str. except for the L preserves the rational cohomology.

$\Rightarrow$ We need to distinguish the class of Kähler manifold for which ω is of integral class (i.e. $\omega \in H^2(X, \mathbb{Z})$)

Note Such cpt complex mfd (X, ω) ($\omega \in H^k(X, \mathbb{Z})$) is called a polarized mfd

$\Rightarrow$ such mfd is necessarily projective.

See §7.1.3 using Kodaira embedding thm.

In this case

$L \in H^k(X, \mathbb{Z})$ and $H^k(X)_{\text{prim}}$ is def on $\mathbb{Z}$.

so Q is integral on $H^k(X, \mathbb{Z})$.

the structure obtained on the primitive part of the cohomology of such a Kähler mfd (X, ω) is then the following.

Def An integral polarized Hodge structure of weight n is given by a Hodge structure $(V_{\mathbb{Z}}, F^p V_{\mathbb{C}})$ of weight n together with an integral intersection form Q on $V_{\mathbb{Z}}$ which is

$$\begin{cases} \text{sym} & \text{if } n \text{ is even} \\ \text{alt} & \text{if } n \text{ is odd} \end{cases}$$

and satisfies conditions i) and ii)

§4 Geometry of Hodge str of weight 1 and 2.

• Weight 1

idea: $\{\text{complex tori}\} \longleftrightarrow \{\text{integral Hodge str of weight 1}\}$.

$\leftarrow$: $V_{\mathbb{Z}}$ Hodge str of weight 1

$V_{\mathbb{Z}} \subset V_{\mathbb{C}}$ can be projected injectively into $V^{0,1}$.

This yields a lattice $V_{\mathbb{Z}} \subset V^{0,1}$ and

$V^{0,1}/V_{\mathbb{Z}}$ is a complex torus

Clearly $V \simeq V' \Rightarrow V^{0,1}/V \simeq V'^{0,1}/V'$.

→: $\mathbb{C}^n/T$: cpx torus.

$\mathbb{C}^n$ can be regarded as $T/\mathbb{R}$ endowed with an almost complex structure.

This yields decomposition

$$(T/\mathbb{R})_{\mathbb{C}} = \underbrace{(T/\mathbb{R})^{1,0} \oplus (T/\mathbb{R})^{0,1}}_{\text{eigen spaces on which } \begin{cases} i \in \mathbb{C} \text{ acts mult. by } i \\ -i \in \mathbb{C} \text{ acts mult. by } -i \end{cases}}$$

eigen spaces on which $\begin{cases} i \in \mathbb{C} \text{ acts mult. by } i \\ -i \in \mathbb{C} \text{ acts mult. by } -i \end{cases}$

Moreover we obtain

$\{\text{abelian varieties}\} \leftrightarrow \{\text{polarized integral Hodge str of weight 1}\}$

For the case of X : kähler mfd.

$$T = H^{0,1}(X)/H^{1,0}(X, \mathbb{Z}) = \text{Pic}^0(X) \quad \text{Picard var.}$$

$$:= \ker \{C_1: H^1(X, \mathbb{Q}^*) \rightarrow H^2(X, \mathbb{Z})\}$$

Fact X : polarized mfd

For degree reason $H^1(X, \mathbb{Q})_{\text{prim}} = H^1(X, \mathbb{Q})$.

and thus the alternating pairing.

$$Q(\alpha, \beta) = \int_X \alpha \wedge \beta \wedge \omega^{n-1} \quad n = \dim X$$

is indeed a polarization

⇒ $\text{Pic}^0(X)$ for proj sm var X is an abelian var.

The most important application of Hodge str of weight 1 is
Griffiths-Torelli Thm.

Thm Two smooth cpx complex curve C and C' are isom.
iff there is an isom $H^1(C, \mathbb{Z}) \simeq H^1(C', \mathbb{Z})$ of integral
Hodge str respecting the polarization.

Lem 7.23

If ϕ is morphism of Hodge str of type (r, r)
 then ϕ is strict for the Hodge str
 i.e. $\text{Im } \phi \cap F^{k+r} W_{\mathbb{C}} = \phi(F^k V_{\mathbb{C}})$

Using this we can define Hodge str on $\text{Im } \phi, \ker \phi, \text{Coker } \phi$

$$\bullet \text{Im } \phi = \bigoplus_{p+q=n} (\text{Im } \phi)^{p, r, q, r}$$

" $\leftarrow$ strict.
 $\phi(V^{p, q})$

$$\bullet \ker \phi = K_{\mathbb{R}} \subset V_{\mathbb{R}}$$

$K_{\mathbb{C}} = \ker \phi \subset V_{\mathbb{C}}$ Then $(K_{\mathbb{R}}, F^k K_{\mathbb{C}})$ is Hodge decomp.

$$\bullet \text{Coker } \phi.$$

So the case of morphisms of rational Hodge str of type $(0, 0)$, the category of rational Hodge str of given weight is an Abelian Category (or sub field of $\mathbb{C}$)

Moreover, The polarised case form an even more rigid category (semi-simple)

Lem 7.26

$V_{\mathbb{Q}} \subset W_{\mathbb{Q}}$ rational sub-Hodge str.
 If the Hodge str of W is polarised, the same holds for the Hodge str on V . and we have a decomp

$$W_{\mathbb{Q}} = V_{\mathbb{Q}} \oplus V'_{\mathbb{Q}}$$

where $V'_{\mathbb{Q}}$ is also a sub-Hodge str of $W_{\mathbb{Q}}$

pf see text, $V'_{\mathbb{Q}} = V_{\mathbb{Q}}^{\perp}$ w.r.t Q